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Digital Design

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PROJECT REPORT

Traffic Light

Student: Hena Šehović

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1. Introduction

1.1 History of Traffic Light

The concept of traffic control dates to the 19th century when the first manually operated gas-lit traffic lights were installed outside the British Houses of Parliament in London in 1868. These early traffic lights were manually operated by a police officer and used semaphore arms during the day and red and green gas lamps at night to control traffic flow. However, their use was short-lived due to a gas explosion that caused injuries, highlighting the dangers of such an approach.

The advent of electricity revolutionized traffic control. In 1914, Cleveland, Ohio, saw the introduction of the first modern electric traffic lights. These lights featured red and green signals and were controlled by a manual switch. This system laid the foundation for automated traffic management. By the 1920s, traffic lights began incorporating yellow signals to indicate a transition between stop and go, further enhancing safety.

Over the decades, traffic lights have evolved significantly. Modern systems now employ advanced technologies such as sensors, timers, and adaptive algorithms to manage traffic flow efficiently. These innovations allow real-time adjustments based on traffic conditions, reducing congestion and improving safety at intersections. Today, traffic lights are a ubiquitous feature of urban infrastructure, critical to maintaining order and preventing accidents on increasingly busy roads.

Picture 1. Traditional Traffic Light

1.2 How Traffic Lights Work

Traffic lights are essential devices for regulating the movement of vehicles and pedestrians at intersections. They typically consist of three signals that convey important instructions through colours and symbols, such as arrows or bicycle icons. The standard sequence of colours includes red to signal stopping, yellow for caution or transition, and green to allow movement. These signals are usually arranged vertically or horizontally in the same order.

While this sequence is standardized globally, variations in specific traffic light rules and operations exist at national and local levels. Modern systems often utilize timers or adaptive control mechanisms, relying on sensors to adjust timing dynamically for improved efficiency and safety. Their operation is fundamentally based on digital logic, involving sequential circuits and finite state machines.

Two-Way Traffic Lights manage the flow of vehicles from two intersecting directions, ensuring orderly and safe movement. The system operates on a sequence that alternates the right-of-way between the two directions, with states for red, yellow, and green lights. Timers are integrated to control the duration of each state.

In a typical setup, the system starts in a state where one direction has a green light and the other has a red light. After a predetermined interval, the green light transitions to yellow, signalling caution. Once the yellow light interval ends, it switches to red, and the opposite direction receives the green light. This cycle repeats, maintaining a balance of traffic flow and safety. Transitions between these states are governed by a state table or state diagram, which ensures that no conflicting signals occur.

1.3 Overview of the Project

This project simulates a simple traffic light system using digital design principles. Designed in Logisim, the project employs a combination of logic gates, flip-flops, and combinational circuits to replicate the functionality of a real-world traffic light. The primary objective is to demonstrate an understanding of sequential circuit design, timing mechanisms, and the integration of multiple components into a cohesive system. By working in a simulation environment, the project provides a practical and visual approach to grasping complex digital design concepts.

Picture 2. Logisim

1.4 The Growing Up of the Idea

The idea for this project began with the need to design a basic, functional traffic light system capable of controlling two-way movement at an intersection. Initially, the concept was inspired by the widespread use of traffic light systems in real-world urban planning to ensure smooth traffic flow, prevent accidents, and promote safety.

At first, the design was simple, focusing on controlling the flow of traffic for each direction separately. However, as I began to explore the problem in more depth, I recognized the importance of sequencing the lights properly to avoid collisions and ensure that each direction gets an equal amount of time for safe passage. This led me to think about how traffic lights need to be synchronized and how timing and state transitions can be modelled digitally.

Incorporating the two-way traffic aspect into the design required an approach that handled alternating green and red lights for both directions while also ensuring that no conflicts arise (such as both directions being green at the same time).

Throughout the development process, I tested different approaches, and ensured the system met the necessary requirements, such as efficient timing, proper signalling, and smooth transitions between different light states. The project gradually evolved from a simple idea into a more complex digital representation of a real-world system, offering a clear solution to manage traffic flow efficiently.

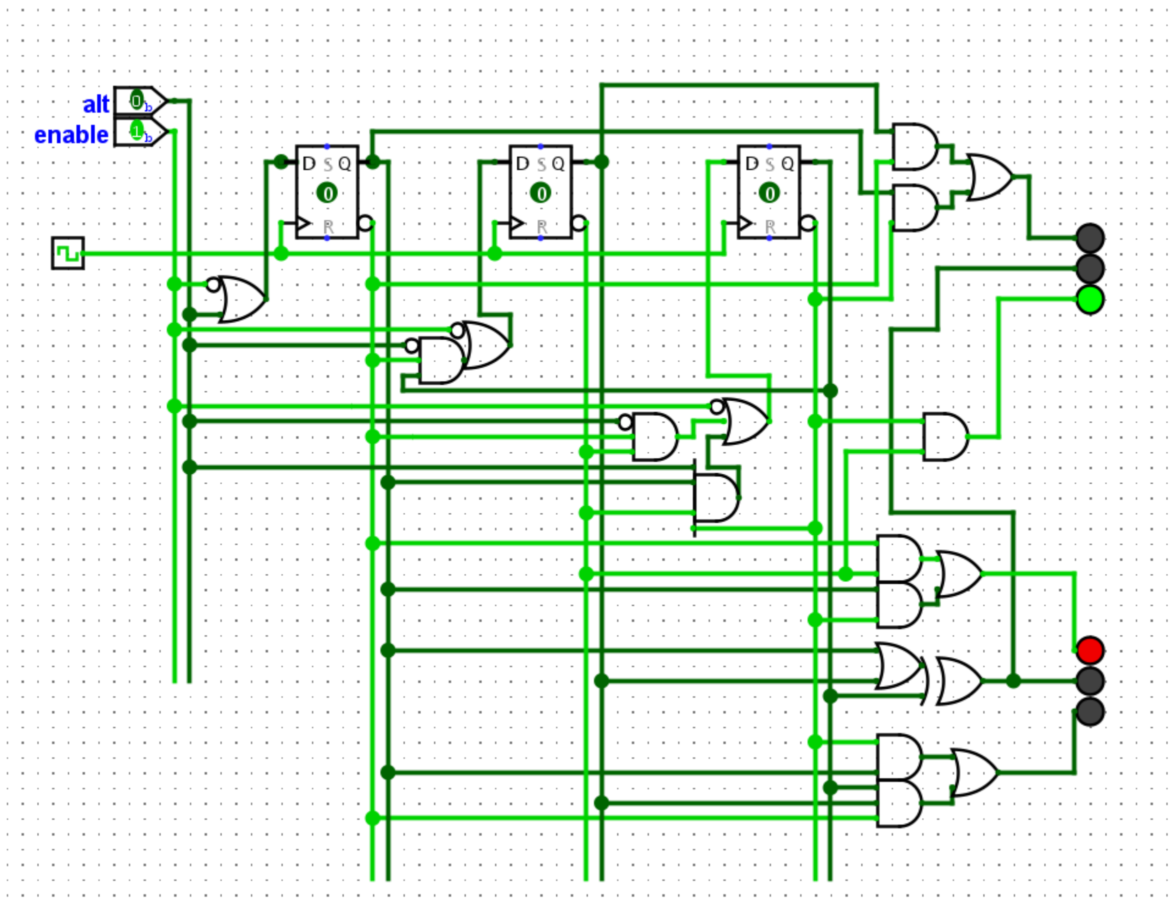
2. The Project

2.1 Visual Representation

The visual representation of the traffic light system is provided to give a clearer understanding of its structure and operation. Below is a description of the key elements and how they are visually represented in the project designed in Logisim.

The overall design consists of one traffic light system, with 2 traffic lights one for each direction of movement. Each traffic light is visually represented using a combination of LEDs that simulate the red, yellow, and green signals. The LEDs are connected to the output of the state machines, and their colours change based on the state of the system. A screenshot of the Logisim circuit shows how all components are connected, including the logic gates, flip-flops, timers, and traffic lights. The visual representation helps to understand how the inputs and outputs work together to create a synchronized system that manages the flow of traffic in both directions.

The circuit was designed with a clear focus on efficiency and simplicity, ensuring that the traffic light system functions correctly while minimizing unnecessary complexity in the design. The visual representation of the project serves as a valuable tool for understanding the underlying logic and operation of the system.



Picture 3. Traffic Light in Logisim

2.2 Project Logic/ Until Realization

The logic behind the traffic light system was developed in several phases, each aimed at addressing specific design challenges while ensuring the project met its core objectives of controlling two-way traffic flow efficiently.

2.2.1 Initial Planning and Design

The first step was to define the functional requirements of the system. The traffic light system needed to manage two-way traffic with alternating green and red signals for each direction. It was also necessary to account for the transition times between the lights to ensure smooth traffic flow and avoid accidents. The system needed to incorporate a simple timing mechanism for controlling the duration of each light (green, yellow, red) for both directions.

2.2.2 State Machine Design

The core of the project is the state machine that controls the behaviour of the traffic lights. I designed one machine that is connected differently to each of the traffic lights. State machine has six main states:

A. None of the lights are on
B. Red light is on
C. Both yellow and red light are on
D. Green light is on
E. Yellow light is on
F. All three Lights are on

Description	State	Code	Red	Yellow	Green
Starting state Enable=0	**A**	111	0	0	0
System working “normal”	**B**	000	1	0	0
	C	001	1	1	0
	D	011	0	0	1
	E	010	0	1	0
System working “alternatively”	**F**	100	1	1	1
	G	101	0	0	0

Table 1. States of the Traffic Light

The state machine uses flip-flops to store the current state of the light and logic gates to determine the next state based on the input from the timer. This allowed the traffic lights to transition through their cycles automatically, based on a defined timing sequence.

The two traffic lights had to be connected to the system, in a way to provide these four states (not including enable and alternative):

1. Green for Direction 1, Red for Direction 2.
2. Yellow for Direction 1, Yellow and red for Direction 2.
3. Red for Direction 1, Green for Direction 2.
4. Yellow and red for Direction 1, Yellow for Direction 2.

2.2.3 Timing Control

The key challenge in this project was determining the appropriate duration for each light colour. To manage this, I used a timer circuit in Logisim to control the duration of each phase. The timer was set to provide an adequate amount of time for each light (green, yellow, red), ensuring that each direction had enough time to pass without creating conflicts.

The timer would trigger state transitions at the end of each light phase, advancing the state machine to the next state in the cycle.

A. Red light is on for 20 seconds
B. Both yellow and red light are on for 4 seconds
C. Green light is on for 20 seconds
D. Yellow light is on for 4 seconds

To simplify our timing and calculations, we set the clock period T=4 seconds. This clock period determines how we measure the duration of each state in terms of clock periods, as follows:

A. Red light is on for 5 periods
B. Both yellow and red light are on for 1 period
C. Green light is on for 5 periods
D. Yellow light is on for 1 period

When we sum up the periods for all states, the total cycle duration of the traffic light system is 12 periods. Each period corresponds to a distinct state in the finite-state machine controlling the traffic light.

2.2.4 Synchronization of Two-Way Traffic

One of the most important aspects of this project was ensuring that the two traffic lights operated in a synchronized manner to avoid conflicts. Both traffic lights needed to be controlled differently, but in a way that ensured each direction would have its own green light at alternating intervals. This was achieved by ensuring that when one direction's green light was on, the other direction’s red light was active. The yellow light was used as a transition phase between red and green in each direction.

Q2	Q1	Q0	Red	Yellow	Green
0	0	0	1	0	0
0	0	1	1	1	0
0	1	0	0	1	0
0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	0	0	0
1	1	0	X	x	x
1	1	1	0	0	0

Table 2. States for Traffic Light 1

Logic Expressions for each LED of the Traffic Light 1:

Red=Q2’Q1’+Q2Q0’

Yellow=Q0’Q1+Q0’Q2+Q2’Q1’Q0

Green=Q0’Q2+Q2’Q1Q0

Q2	Q1	Q0	Red	Yellow	Green
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	1	1	0
0	1	1	1	0	0
1	0	0	1	1	1
1	0	1	0	0	0
1	1	0	X	x	x
1	1	1	0	0	0

Table 3. States for Traffic Light 2

Logic Expressions for each LED of the Traffic Light 2:

Red=Q2’Q1+Q0’Q2

Yellow=Q0’Q1+Q0’Q2+Q2’Q1’Q0

Green=Q0’Q1’

2.2.5 Debugging and Optimization

After the initial design was completed, I tested the circuit to ensure the traffic lights were correctly alternating and that no conflicts occurred.

2.3 Components Used

1. Clock Signal: Used to generate a clock pulse that drives the sequential behaviour of the circuit.

2. D Flip-Flops: Three D flip-flops are used for storing the state of the system. These flip-flops are fundamental for sequential logic and state transitions.

3. Logic Gates:

AND Gates: Several AND gates are used to ensure that specific conditions must be met simultaneously for a signal to propagate.

OR Gates: OR gates combine multiple input signals, allowing the activation of lights based on various conditions.

NOT Gates (Inverters): NOT gates invert signals, ensuring proper logical conditions for transitions.

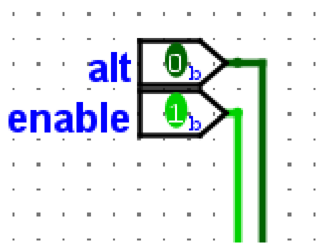
5. LEDs: Three LEDs (Red, Yellow, Green) represent the traffic light outputs. The LEDs visually indicate the current state of the system.

6. Connecting Wires: Wires establish logical connections between the components and ensure proper flow of signals.

7. Input Pins: Input pins are standard Logisim components that act as external control signals, providing binary inputs (0 or 1) to influence the behaviour of the circuit.

2.4 Additional Components

To enhance the functionality of the traffic light system, two new components were introduced: **Enable** and **Alternative**. These components add further control and adaptability to the system:



Picture 4. Additional Components

2.3.1 Enable Component

The Enable component acts as a global switch for the traffic lights. When Enable is set to 0, the traffic light system is entirely disabled, and all traffic lights remain off. When Enable is set to 1, the traffic light system functions normally according to the pre-defined state transitions and timing sequences.

2.3.2 Alternative Component

The Alternative component introduces a special operating mode for the traffic light system. When Alternative is set to 0, the traffic lights operate in their regular sequence, alternating between green, yellow, and red states for both directions. When Alternative is set to 1, the traffic lights enter a blinking state, where all lights blink simultaneously. This blinking state is designed to simulate conditions where normal operation is interrupted due to a malfunction, emergency, or specific traffic control requirements.

These additional components significantly enhance the system's versatility by allowing it to adapt to various operational scenarios. The Enable switch ensures that the system can be fully deactivated, when necessary, while the Alternative mode provides a visual indication of abnormal conditions or special operational requirements.

2.5 Combining Components

The integration of all components was a critical phase in ensuring the functionality of the traffic light system. Each component, from the state machine to the timer and logic gates, was designed to work in harmony to control the two-way traffic efficiently. In Logisim, the components were connected as follows:

- The clock generator feeds the D flip-flops, enabling them to toggle their states at regular intervals. This setup ensured that the traffic lights transitioned through the required cycles without conflicts.
- The outputs of the flip-flops are connected to logic gates, which implement the conditions for lighting up red, yellow, or green LEDs on both traffic lights.
- The LEDs are connected to the output of these gates, visually representing the state of the traffic light.
- The Enable input pin was integrated to act as a global controller for the system and the Alt (Alternative) input pin was added to introduce an additional mode of operation.

After the components were connected, the system underwent testing. The primary goal was to ensure that all transitions occurred as expected, with no conflicts between the two directions. The timer and state machine were carefully monitored to verify the timing sequences, and any issues with synchronization were addressed. Additionally, the Enable and Alt functionalities were tested to confirm proper switching between modes and system-wide control.

3. Conclusion

The traffic light system project successfully achieved its core objectives of managing two-way traffic flow through a controlled and synchronized mechanism. By leveraging key components such as the state machine, timer, flip-flops, and logic gates, the system was able to transition between four essential states, ensuring smooth and safe traffic flow. The use of a defined clock period and careful timing control allowed for precise management of the green, yellow, and red-light phases. Additionally, the synchronization of the two traffic lights ensured that conflicts were avoided, with each direction receiving its green light at alternating intervals. The addition of the Enable and Alternative components enhanced the functionality of the system. The Enable component provided a global control mechanism to turn the system on or off, while the Alternative component introduced a blinking mode for emergency or malfunction scenarios. These features added flexibility and robustness, making the system adaptable to different operational conditions.

The testing and debugging phase confirmed the system's efficiency, reliability, and versatility. Any potential issues were promptly addressed to ensure accurate timing, synchronization, and functional correctness. Overall, the project demonstrated the successful application of basic digital components and advanced control mechanisms to a real-world problem, highlighting the importance of system integration and adaptability.

4. Result

The implemented traffic light system operated as intended, with both traffic lights alternating through the defined states in a synchronized manner. The system managed to control two-way traffic flow without any conflicts, ensuring that each direction received its green light in turn while the other direction was held at red. The inclusion of the Enable component ensured that the traffic lights could be deactivated when required, adding an additional layer of control to the system. The Alternative component allowed the traffic lights to blink in unison when activated, simulating emergency or malfunction scenarios. These new features were successfully integrated and tested alongside the core functionalities of the system.

The state transitions were smooth and automatic, driven by the timer circuit and managed by the state machine. The total cycle duration of 12 clock periods ensured that each light phase had adequate time, and no delays or unsafe conditions were observed. The system was tested under various scenarios, and it successfully handled the timing requirements for each light colour, confirming its practical viability. The project concluded with a functional, reliable, and efficient solution to controlling two-way traffic flow, demonstrating the effective use of digital logic and timing control in traffic management.

5. Resources

https://en.wikipedia.org/wiki/Traffic_light

List of figures:

Picture 1. Picture of the Traffic Light

https://cdn11.bigcommerce.com/s-8zoy5b/images/stencil/original/products/266/1831/Signaworks-TL12-3-12-Inch-Diameter-Lens-LED-Traffic-Light-Signal-3-Color-12-24VDC-Ready-To-Wire-Angled-On-11-2023__11661.1700156585.png

Picture 2. Logisim Logo

https://dashboard.snapcraft.io/site_media/appmedia/2023/03/logisim-icon-512.png

Picture 3.

Picture 4.

Table 1.

Table 2.

Table 3.